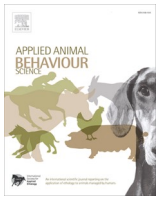
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Dairy farmers' attitudes and empathy and cattle's reactions to humans in cow-calf contact systems compared to early separation

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ABSTRACT

A good relationship between stockpeople and non-human animals (NHA) has positive effects on the welfare of different species such as cattle through the sequential relationship of human attitudes, human behaviour and, subsequently, NHA behaviour and welfare. Cow-calf contact (CCC) rearing of dairy calves can enable welfare benefits and is increasingly gaining farmers' interest, often motivated by the possibility to offer cattle the opportunity for more natural behaviours. Thus, CCC farmers' attitudes towards cattle and their relationship with them may differ from those who practise early separation (ES) of calf and cow. Moreover, CCC rearing bears the risk for reduced human-calf contact when bucket feeding ceases. Thus, we studied the human-cattle-relationship on 25 dairy farms that practised CCC and 25 that practised ES of cow and calf, i.e. within 24 h after birth. Farmers' (n = 57 each for CCC and ES) attitudes towards cows and handling them, being predictors for their handling behaviour and a proxy for the relationship with cattle, as well as farmers' empathy with NHA were assessed using questionnaires. The avoidance distance (AD) of calves, heifers, and cows, reflecting the cattle's relationship to humans, was assessed. Principal component analysis revealed 14 attitude and one empathy component. Potential differences between CCC and ES farmers' attitudes and empathy were evaluated by linear mixed models. Linear regression models were fitted to investigate potential differences between rearing systems in the percentage of heifers or cows that could be touched, the median AD, and the percentage with an AD > 50 cm. For calves' AD, a linear mixed model was used to account for different test locations. Cow-calf contact farmers agreed more on *cows' needs, liked positive contact* with cows more and tended to agree less on *impatient behaviour* during moving than ES farmers. Empathy correlated with almost all attitude components. There was no difference in the AD of CCC and ES calves and cows, but CCC heifers had higher ADs than ES heifers. Several attitude components were predictors for the cattle's AD. The results indicate more positive attitudes among CCC farmers, no general effect of CCC on the AD, and associations between attitudes and AD.

1. Introduction

On dairy farms, separation of cow and calf right after birth is routinely practiced despite adverse effects on cow and calf welfare (EFSA Panel on Animal Health and Animal Welfare et al., 2023). For example, prolonged cow-calf contact (CCC) reduces the risk for abnormal oral behaviours in calves and improves their social competence (Fröberg and Lidfors, 2009; Roth et al., 2009; Wagner et al., 2012, 2015). In cows, CCC can improve udder health (Fröberg et al., 2008). Moreover, CCC allows cows to perform maternal behaviour, for which they are highly motivated, and which can thus be considered to positively affect their well-being (von Keyserlingk and Weary, 2007; Wenker et al., 2020). An increased awareness of the benefits of CCC and the disadvantages of early separation (ES) has also led to greater interest among farmers and application in practice (Barth et al., 2021; Placzek et al., 2021).

Caretakers and decision-makers are the main factor for the well-being of non-human animals (in the following: animals) used for

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farming purposes (reviewed in Waiblinger et al., 2025). Their relationship with animals under their care significantly influences handling practices and management decisions that directly or indirectly determine animal welfare (reviewed in Waiblinger et al., 2006; Hemsworth and Coleman, 2011; Waiblinger, 2019). The human-animal relationship (HAR) is "the degree of relatedness or distance between the animal and the human, i.e., the mutual perception" (Waiblinger et al., 2006). The relationship from the animal's perspective (animal-human relationship, AHR) is determined by the relative strength of positive and negative emotions induced by human-animal interactions (HAI), and depends on frequency, duration, and type of human contact, including especially movements and vocal and physical interactions, and the consistency of the interactions (Boivin et al., 2003; reviewed in Waiblinger et al., 2006).

According to the Theory of Planned Behaviour (Ajzen and Fishbein, 1980), the most important factors influencing a person's behaviour are attitudes, that are thus an integral part of the HAR in farmers; personality, empathy, knowledge, experience, the actual situation, perceived behavioural control and peer pressure are also relevant (Seabrook, 1984; Coleman et al., 1998; Waiblinger et al., 2002; Hemsworth, 2003; reviewed in Hemsworth and Coleman, 2011; Waiblinger, 2019). Attitudes are "a psychological tendency that is expressed by evaluating a particular entity with some degree of favour or disfavour" (Eagly and Chaiken, 1993; reviewed in Hemsworth and Coleman, 2011), with behavioural attitudes referring to a specific behaviour, in our context handling practices towards cattle. Behavioural attitudes are best for predicting behaviour, while general attitudes about cattle, i.e., beliefs about cattle, influence behaviour mostly indirectly via effects on behavioural attitudes (Waiblinger et al. 2002). Training of stockpeople based on cognitive-behavioural interventions efficiently changed the welfare of dairy cows through changes in stockpeople's attitudes and, subsequently, behaviour towards the cows, and more positive attitudes towards cows were associated with calmer cows during handling procedures (Hemsworth et al., 2002; Ebinghaus et al., 2018). Empathy (i.e., the capacity to vicariously experience the emotions of another (Coleman et al., 1998)) is seen as an antecedent to attitudes (Ajzen and Fishbein, 1980; reviewed in Hemsworth and Coleman, 2011), but no published papers have investigated potential links between behavioural attitudes and empathy in dairy farmers yet.

Farmers practising CCC often highlight the importance of allowing their cattle to perform natural behaviour, that calves need their mothers, and that cow and calf are "happier" if they are together as their motivation for CCC (Johanssen et al., 2023; Vaarst et al., 2020; Waiblinger, 2025). This indicates generally positive attitudes of CCC farmers towards (the needs of) cattle. However, attitudes of CCC compared to ES farmers regarding beliefs about cattle and on handling them, i.e., attitudes that are predictors for farmer behaviour and reflect their relationship with cattle, have not yet been studied.

In addition, (whole-day) CCC systems bear the risk for the development of a poorer HAR in calves (Waiblinger et al., 2020) as they must not be bucket-fed (which is still common for ES calves in Austria (Klein-Jöbstl et al., 2024), so that no regular HAI must take place during feeding times. Also, some CCC farmers reported their calves to be more "wild" compared to when they were practising ES (Vaarst et al., 2020; Waiblinger and Hebesberger, 2025). There is no comprehensive evidence from on-farm studies on the HAR in calves, heifers or cows in CCC systems so far.

In this study, we examined the HAR on CCC or ES farms: farmer's attitudes towards (handling of) cows and their empathy with animals were evaluated via questionnaires, and the avoidance distance of calves, heifers and cows towards humans was assessed. We hypothesized that (i) CCC farmers have more positive attitudes and higher empathy than ES farmers and that (ii) there are no differences in the avoidance distance of calves, heifers or cows between CCC and ES farms. The study is part of the transdisciplinary project "COwLEARNING for sustainable beef and dairy supply" (COwLEARNING, 2026). Besides the HAR, welfare of cattle, farmers' work satisfaction, their interconnection and further aspects of ecological, economic and social sustainability were assessed and are or will be presented elsewhere (e.g. Rademann et al., 2025, 2026, submitted).

2. Methods

2.1. Ethics Statement

The project was approved by the Ethics and Animal Welfare Committee of the University of Veterinary Medicine, Vienna in accordance with the University's guidelines for Good Scientific Practice (ETK-11/01/2023). Due to the observational nature, the study is not classified as animal experiments under Directive 2010/63/EU or the Austrian Act on Animal Experiments. All participating farmers received extensive information about the project and agreed that their data may be used (anonymized) for scientific research and publication.

2.2. Study design and farms

Fifty dairy farms, of which 25 practised prolonged cow-calf contact (CCC, median: 12 weeks, two farms < 12 weeks, seven farms > 12 and ≤ 16 weeks, and three farms > 16 weeks) and 25 separated cow and calf within 24 h after birth (early separation, ES), were visited from February-April 2023 and October-April 2024, i.e., outside pasture season, for one day each. Cow-calf contact and ES farms were approximately balanced for herd size and geographical region within Austria (Table S1). Farm characteristics are presented in Table S1. For a detailed description of the selection and data collection process, see Rademann et al., (2025).

2.3. Assessing the human-animal relationship

2.3.1. Attitudes and empathy

For assessing the farmers' relationship with their cattle and their empathy with animals, all persons caring for the cattle filled in a questionnaire on attitudes including a section on farmers' demographics that included age, gender, years of experience with cattle (if yes, which type), whether they had grown up with cattle or on a farm (if yes, which type) and whether they had an agricultural education (if yes, which type), as well as a questionnaire on empathy.

Questionnaires were filled in on paper. Farmers were instructed orally before filling in the questionnaire, and the instructions were also written on the front page. Instructions included that the questions were to be answered alone to collect the farmer's personal opinions, that data would be handled anonymously, that there are no right or wrong answers, that questions should be answered spontaneously, and that every question should be answered. Questionnaires were labelled with an abbreviation for each farm and person, so names were not collected.

2.3.1.1. Questionnaire on Attitudes. The questionnaire developed and validated for assessing the HAR (Waiblinger et al., 2002, based on Hemsworth et al., 2000) that has been used in further projects in the past (Rouha-Mülleder et al., 2009; Ivemeyer et al., 2018; Andreasen et al., 2020), was employed. It comprised 111 items that were allocated to four main sections (one with two subsections): 1) general attitudes towards, i.e. beliefs about, cows (27 items) and calves (three items), 2) behavioural attitudes regarding handling of cows when moving them (29 items), and during milking (14 items), 3) beliefs about importance of regular contact with cows (25 items) and 4) affective attitudes, i.e. (dis) liking working with and special contact with cows (13 items). Each item was to be rated on a 7-point Likert Scale. The scale ranged, depending on the section, from 1 (= strongly disagree) to 7 (= strongly agree), 1 (= not at all important) to 7 (= very important), or 1 (= do not like it at all) to 7 (= like it very much). Beliefs about cows and calves are referred to as

general attitudes as they do not deal with behaviour towards cattle (Waiblinger et al., 2002; reviewed in Hemsworth and Coleman, 2011). Behavioural attitudes comprise beliefs and intentions about handling practices of cattle in the milking parlour and in general, and affective attitudes refer to the emotional responses of a person related to contact with cows in different situations (Waiblinger et al., 2002; reviewed in Hemsworth and Coleman, 2011). Example questions for each section of the questionnaire are provided in Table S2.

2.3.1.2. Questionnaire on Empathy. The questionnaire comprised 27 items (both positive and negative statements) that were to be rated on a Likert Scale from 1 (= strongly disagree) to 7 (= strongly agree). It was developed based on earlier research that investigated the relationships between empathy, attitudes, and caretaker behaviour (Coleman et al., 1998; Coleman, 2004; Hanna et al., 2009) and has been used previously (Mülleder and Waiblinger, 2004). The questionnaire included statements relating to the attribution of emotions to animals and reflecting the respondent's own emotions to animals. Example questions are provided in Table S3.

2.3.2. Human-cattle contact

In addition to the questionnaire on attitudes, a management interview was conducted with the farm manager that included questions on human-cattle contact, i.e., the number of milkers, the number of times being in the herd each day, the amount of time per day that farmers spend in the barn (i.e., in contact with the cattle), how often the cattle were brushed by humans and how many of the cows the farmers can recognise individually without checking the ear tag or collar. Regarding contact to humans of the calves, ES calves were always handled right after birth due to the separation from their dam and manual colostrum feeding. This continued throughout the entire rearing period as almost all calves were manually bucket-fed (only one farm had an automatic milk feeding system after bucket-feeding the calves in the first week of life). Also, CCC calves were routinely handled on most CCC farms, as 19 of the 25 CCC farms fed additional colostrum by hand, and almost half of the farms were farms with part-time contact where calves were usually handled several times a day during contact times.

2.3.3. Avoidance distance tests

To assess the cattle's relationship with humans, the avoidance distance (AD, Windschnurer et al., 2009; Lürzel et al., 2015) was measured. To conduct the test, the experimenter placed herself 2 m in front of the cattle to be tested and ensured that the cattle had noticed her. She then approached the cattle by walking in a standardised speed of one step per second with her arm stretched outward toward the cattle in an angle of 45°, the back of the hand pointing to the front. In the moment that a withdrawal movement, i.e., a movement to the back or to the side, was noticed, the distance between the experimenter's hand and the cattle's muzzle at the time of withdrawal was estimated and noted down in 10 cm resolution. In case the cattle could be touched, "0" was noted down. Heifers and cows were always tested at the feeding place. As calves were sometimes single-housed, sometimes group-housed with and sometimes group-housed without feeding racks, the AD tests had to be conducted at three different test locations for calves: 1) from outside when single-housed, 2) at the feeding place, i.e., in group-housing with a feeding rack, and 3) in the barn, i.e. in group-housing without a feeding rack. On CCC farms with whole-day contact, calves were tested with dams or foster cows present in the barn. On CCC farms with part-time 2x daily contact, AD tests were performed outside of the contact period, as calves were too engaged in suckling and social contact during the contact time so that testing "without disturbance" was not reliably possible. The AD tests were part of a welfare assessment using Welfare Quality® protocols (Welfare Quality® Network, 2009; Gratzer et al., 2010). The AD tests were always performed at the beginning of the assessment, and each cattle was tested once. For a detailed description of the entire data collection, see Schneider et al. (2024) and Rademann et al. (2025). Calves were defined as a bovine animal up to six months old according to Regulation 2008/119/EC, and heifers were defined as female cattle aged $>$ 6 months who had not yet given birth to a calf (Council of the European Union, 2008; Gratzer et al., 2010).

2.4. Statistical Analysis

2.4.1. Attitudes and empathy

To identify attitude components, principal component analyses (PCA) with Varimax rotation, based on correlation matrices, were run separately for each section of the attitude questionnaire, i.e. five PCA in total. In addition, a principal component analysis was conducted using all the items from the questionnaire on empathy. All PCA followed the procedure of previous studies, e.g., Waiblinger et al. (2002) and Mersmann et al., (2022). The number of components per section was determined using a scree plot, and all components were required to have an eigenvalue $>$ 1 (Tabachnik and Fidell, 2013). The suitability was checked using the Kaiser-Meyer-Olkin criterion ($>$ 0.6) and Bartlett's test of sphericity ($p < 0.05$, Arsham and Lovric, 2011). Missing values were excluded pairwise. An item was included in a component if its loading exceeded 0.3 and did not load on any other component, or if the loading exceeded 0.6 and the loading on another component was $<$ 0.4. Principal component analysis revealed four components for the section on general attitudes, two or four components for behavioural attitudes towards moving or milking cows, respectively, one component for beliefs about the importance of contact, and three components for affective attitudes, i.e. 14 attitude components in total (Table S4). The analysis of the empathy questionnaire confirmed one component for empathy (Table S5). Due to some missing items, components on behavioural attitudes and beliefs about contact are only available for 56 ("use of kickbows": 55) CCC and 57 ES farmers, and components about affective attitudes are only available for 56 CCC and 56 ES farmers.

Cronbach's Alpha (ten attitude and the empathy component $\geq$ 0.7, four attitude components $>$ 0.6, DATAtab, 2025) was used to assess the reliability of each component. For further analysis, component values were computed by calculating a composite score, i.e. the mean value of the items within each component. In case of a negative loading of an item, the item was included after recoding, i.e., by subtracting the item value from the maximum value of the scale plus 1 (e.g. in case of a 7-point Likert Scale, the value 2 was recoded to 6: (7 +1) - 2).

To determine potential differences in attitudes or empathy between the farmers of the two rearing systems, linear mixed models (LMM) were calculated for each of the 14 attitude and the empathy component, i.e. 15 models in total. Rearing system (i.e., CCC or ES) was always included as fixed factor and farm as random factor. In addition, demographic variables potentially influencing attitudes or empathy were pre-selected to be included as independent variable in the LMM based on univariate analysis, either by Spearman rank correlation in case of metric variables, or Mann-Whitney U Tests in case of dichotomous variables. If a variable was associated significantly ($p \leq 0.05$) or tended to ($p \leq 0.1$) with the respective component, it was included in the model as an independent variable (Table S6 and S7). The independent variables included in each model are shown in the results section. No model reduction was performed. Due to the pre-selection of independent variables, p-values must be interpreted with caution. We also calculated models containing all the potential independent variables (i.e., without pre-selection), and the results regarding rearing system as our variable of interest were essentially the same. However, we decided for the models with pre-selection due to better model stability. Associations between potential influencing variables were calculated as well, with the described tests depending on distribution of the variable; Chi^2 Tests were used for two categorial variables (Table S8 and S9). The Variance Inflation Factor (VIF) was used to assess multicollinearity: any variable with a VIF $\geq$ 5 would have been excluded; however, no such variable needed to be excluded. Model assumptions (homoscedasticity and normal distribution of residuals)

were checked graphically.

To determine potential associations between empathy and attitude components and among attitude components, Spearman rank correlation coefficients were calculated.

For further analysis, we calculated different farm values for empathy and attitudes by calculating the mean of the composite score of each attitude and empathy component for (i) all the people working with the cattle on the farm (i.e. farm mean), (ii) all people caring for the calves (i.e. calf caretaker) or decision makers (usually one person, but sometimes more than one; i.e. farm manager).

2.4.2. Avoidance distances

For the analysis of the AD as a proxy for the relationship to humans from the cattle's perspective, the percentage of calves, heifers, or cows who could be touched (= AD0), the median avoidance distance per calf, heifer, or cow, and the percentage of calves, heifers, or cows with an AD of > 50 cm (= AD50) per farm were calculated. To account for potential differences between rearing systems in farmers' attitudes or other factors potentially influencing the cattle-human relationship, Spearman rank correlation coefficients were calculated between AD0, median AD and AD50 and farmers' attitudes and empathy components (Table S10), the variables of human-cattle contact (i.e. the number of milkers, how often cows were brushed by the farmers, the amount of cows the farmer recognised individually, the number of times the farmers walked through the herd each day, the time the farmers spent in the stable per day), and the number of calves, heifers, or cows, respectively (Table S10 and S11). Regarding attitudes and empathy, correlations were calculated for 1) the farm manager mean and 2) the calf caretaker mean (for calves) or the farm mean (for heifers and cows). To test for rearing system differences and potential influences of attitudes, empathy, or human-cattle contact, linear regression models (method: enter) with AD0, median AD, and AD50 of heifers, and cows, respectively, as target variables were calculated. To account for differences in test locations among calves, a LMM with the individual calves' AD as the target variable was fitted instead. Rearing system and farm were included as independent variables in all linear regression models. In the LMM, rearing system and age category (0–2 months or 3–6 months) were included as fixed and farm as random factor. In addition, the variables that were associated significantly ($p \leq 0.05$) or tended to be associated ($p \leq 0.1$) with one of the AD measures for each category (i.e. calf, heifer, or cow) in the univariate analysis were included as independent variables. The Variance Inflation Factor (VIF) was used to assess multicollinearity: any variable with a VIF ≥ 5 would have been excluded; however, no such variable needed to be excluded. Model assumptions (homoscedasticity and normal distribution of residuals) were checked graphically. The composition of each model is shown in the results section.

3. Results

3.1. Participating farmers and human-cattle contact

We received 57 questionnaires from CCC farmers and 57 questionnaires from ES farmers, i.e. 114 questionnaires in total. Table S1 shows the descriptive results of the demographic variables (age, years of experience in working with cattle, agricultural education, and grown up on the farm) of the farmers. On both CCC and ES farms, there was a median of two milkers per farm, and it ranged from 0 - 4 on CCC and 0–3 on ES farms (0 indicates an automatic milking system, this was the case at one CCC and two ES farms). On most cow-calf contact and ES farms, the herd was checked twice a day (median: 2 for both rearing systems; range: 2–15 times on CCC and 2–5 times on ES farms), and CCC and ES farmers spent on average 5 h / day in the barn (median: 5 for both rearing systems, range: CCC farmers: 3.17–10 h, ES: 3–8.25 h). Farmers reported to brush their cows at least once a month on four CCC or one ES farm, less than once a month but more than once a year on one CCC or three ES farms, and once a year, less often, or never on 20 CCC or 21 ES farms, respectively. On 23 CCC or 19 ES farms, farmers reported that they could recognise 100% of their cows individually, on one CCC or one ES farm, they reported 95 - < 100% of the cows, and on one CCC or five ES farms, < 95% of the cows. There was no difference between CCC and ES farms for all of these variables.

3.2. Attitudes and empathy

Cow-calf contact farmers agreed more on the *needs of cows*, *liked positive contact* with cows more and tended to agree less on *impatient behaviour during moving* compared to ES farmers (Fig. 1, Table 1). There was no difference between rearing systems for the other components. In general, farmers held mainly positive attitudes towards cows and the behaviour and contact with them, although a greater variation was seen for some components, particularly for behavioural attitudes (Fig. 1, Table 1).

3.2.1. Correlation of empathy with attitudes

Empathy was correlated with 12 of the 14 attitude components (Table 2). The more empathy towards animals the farmers showed, the more positive general, behavioural, and affective attitudes and the less negative behavioural and affective attitudes they held (Table 2).

3.3. Avoidance Distance

The descriptive results of the Avoidance Distance of heifers and cows are shown in Table 3, and Fig. 2 shows CCC and ES calves' AD in the three different test locations. In the LMM, rearing system was not a predictor for calves' AD (Table 4). Calves' AD was largely influenced by test location but also associated with farm managers' agreement on positive characteristics of cows (Table 4, Fig. 2). The behavioural attitudes associated with AD of calves in univariate analysis were not identified as predictors (Table S10, Table 4), probably because they were confounded with the general attitude and, partly, with each other, as reflected by the change in the direction of association of *importance contact* from univariate to multivariate analysis. Regarding test location, AD was lowest if calves were tested in single housing (i.e. on 15 ES but only one CCC farm) compared to calves tested at the feeding place or in the barn, and calves tested at the feeding place had lower ADs compared to calves tested in the barn (Table 4, Fig. 2). Also cows' AD did not differ between rearing systems. The more time farmers spent in contact with their cows, the lower the median AD and the fewer cows in the herd had an AD > 50 cm (Table 5). In addition, cows on farms where the farm manager agreed more on *punishing behaviour during milking* tended to have higher median ADs and a lower percentage of cows that could be touched (Table 5).

However, AD of heifers differed between rearing systems: A lower percentage of heifers on CCC farms could be touched, and both the median and the percentage with an AD > 50 cm were higher in CCC heifers (Table 5). Similar to the results in cows, the percentage of heifers that could be touched was higher the more time the farmers spent in the barn (Table 5). Moreover, AD of heifers was lower on farms with a lower number of heifers, where farmers agreed more on *positive characteristics of cows*, and tended to be higher on farms where farmers agreed more on *punishing behaviour during milking* (Table 5).

Across rearing systems, there was a large variation in the AD of calves and heifers, although the percentage of heifers with an AD > 50 cm was low (< 10%) on most farms (Table 3). Median AD of cows was 0, and most farms had only a limited proportion of cows with an AD > 50 cm (Table 3).

4. Discussion

The results partly confirm our hypothesis that CCC farmers have more positive attitudes and that there is no general difference in the HAR between rearing systems. Cow-calf contact farmers agreed more on

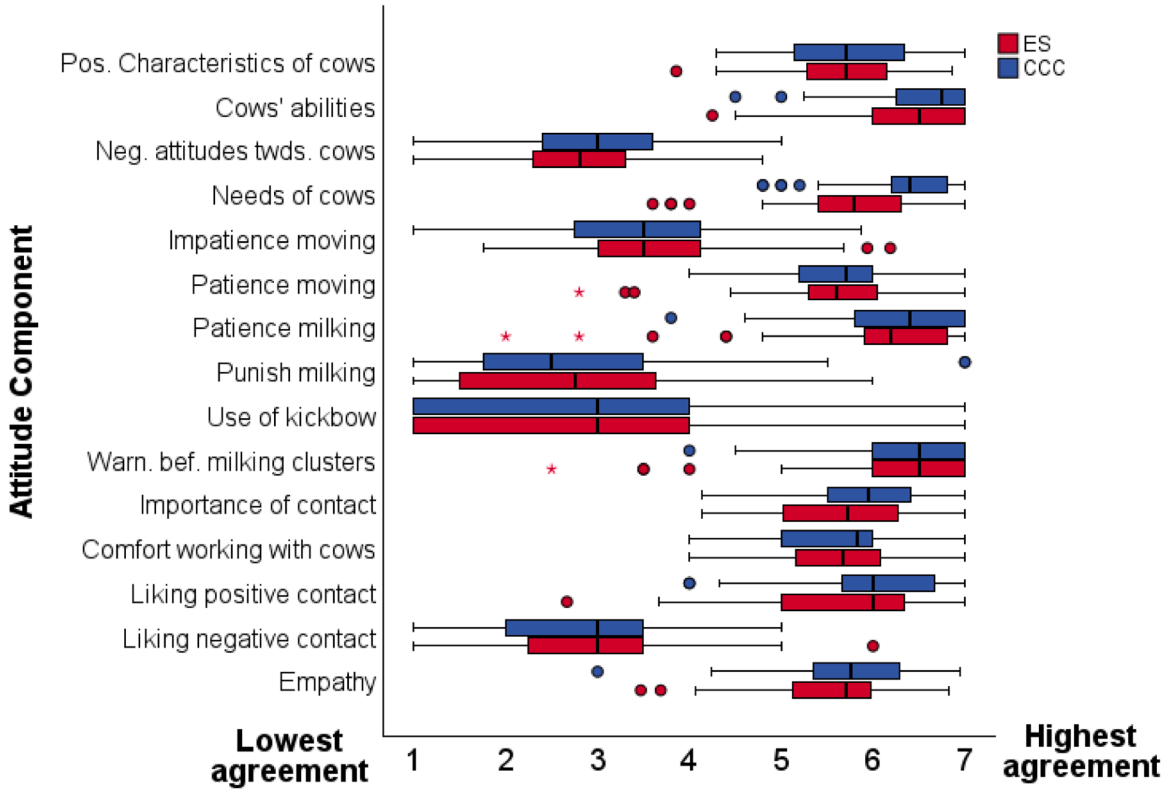


Fig. 1. Descriptive results of the components on attitudes and empathy that were extracted from the questionnaires for n = 57 cow-calf contact (CCC) and n = 57 early separation (ES) farmers from 25 farms each.

cows' needs, *liked positive contact* with cows more and tended to agree less with *impatient behaviour during moving* compared to ES farmers, but no differences were found for the other attitude components. No differences between the AD for calves and cows on CCC or ES farms were found, but CCC heifers had higher ADs. Several attitude components and variables on human-cattle contact were associated with the cattle's ADs.

4.1. Attitudes and empathy

4.1.1. CCC and ES farmers

In line with our hypothesis, cow-calf contact farmers had more positive attitudes towards the *needs of cows* and tended to agree less on *impatient behaviour during moving*. Previous studies revealed that CCC farmers want "cow and calf to have a good time", "to get the best opportunity for growth and animal health", and that "cows are such good mothers" (Neave et al., 2022; Johanssen et al., 2023; Waiblinger, 2025). In addition, 90% of practising CCC farmers and 46% interested in the system chose "enabling their cattle to engage in natural behaviours" as their motivation for CCC, confirming their stronger agreement on cattle's needs (Waiblinger, 2025). One item included in the component '*needs of cows*' specifically asked for the agreement with the statement "cows feel better when they have contact with their calf', and CCC farmers were expected to agree more on this than ES farmers.

Cow-calf contact farmers also reported enjoying working with cows more since practising CCC and that they value actively seeking contact with (mother) cows (Vaarst et al., 2020; Neave et al., 2022; Johanssen et al., 2023; Waiblinger, 2025). This aligns with our results that CCC farmers had more positive attitudes towards *contact with cows*.

4.1.2. Correlations between empathy and attitude

Empathy was correlated with almost all attitude components. This is consistent with a previous study in pig stockpeople in which empathy was correlated with most attitude components but not with stockperson behaviour (Coleman et al., 1998). Thus, following the Theory of Planned Behaviour (Ajzen and Fishbein, 1980), attitudes are seen as the best predictors of behaviour, whereas empathy is an antecedent to attitudes and is therefore important but only indirectly influences behaviour (Coleman et al., 1998; Hemsworth and Coleman, 2011).

4.2. Cattle-Human Relationship: Avoidance Distance

4.2.1. Cow-calf contact and early separation

The lack of a difference of ADs of calves between the two rearing systems is in line with our hypothesis and with a previous study conducted in an experimental setting but on commercial farms (Bieber et al., 2022). In a study conducted on a research farm, however, dam-reared calves had higher ADs compared to ES calves (Waiblinger et al., 2020). Yet, dam-reared calves received only minimal handling compared to ES calves who had regular human contact during feeding times, and the authors also found a decreased AD in the calves that had to be handled more (Waiblinger et al., 2020). Thus, limited contact to humans and not CCC per se led to the difference in the AD. There is an increased risk that calves in (whole-day) CCC systems develop a weaker AHR as no regular HAI that are essential for the formation of the AHR (Lensink et al., 2000; Lürzel et al., 2015) must take place during feeding times, and farmers have reported increased "wildness" in CCC calves (Vaarst et al., 2020; Waiblinger, 2025). Moreover, it has been reported that, as the dam is the main socialisation partner of the calf, practising human-calf interactions may be more difficult in the presence of a cow (Krohn et al., 2003). However, quality and quantity of HAI are mainly determined by farmers' attitudes (Waiblinger and Menke, 1999; Waiblinger et al., 2002; reviewed in Waiblinger et al., 2006; Hemsworth and Coleman, 2011). This is consistent with our results which suggest that appropriate human-calf interactions to form an AHR comparable to ES calves are possible in CCC systems. In addition, especially the first days are of relevance for the formation of the AHR (Jago et al., 1999), and 19

Table 1
Results of the Linear Mixed Models on empathy and attitudes for n = 57 CCC and n = 57 ES stockpeople from 25 CCC and 25 ES farms, respectively. Significant ($p \leq 0.05$) p-values are marked in bold, tendencies ($0.05 < p \leq 0.1$) are marked in *italics*.

Table 1

Dependent Variable	Independent variable	F	p	β	SE	df	Marginal R^b (model)	Conditional R^b (model)
Empathy	Rearing system[a]	1.050	0.312	0.167	0.163	38.139	0.057	0.225
	Education[b]	1.935	0.168	-0.268	0.192	93.458		
	Yrs exp. w. cattle	2.496	0.118	-0.007	0.004	91.559		
Positive Characteristics of cows	Rearing system[a]	0.922	0.342	0.129	0.134	42.535	0.074	0.110
	Age	2.304	0.132	-0.011	0.007	100.387		
	Yrs exp. w. cattle	0.113	0.737	-0.002	0.006	102.331		
Cows' abilities	Rearing system[a]	2.012	0.163	0.186	0.131	47.449	0.066	0.171
	Age	1.140	0.288	-0.007	0.007	97.679		
	Yrs exp. w. cattle	0.333	0.565	-0.003	0.006	100.294		
Negative attitudes towards cows	Rearing system[a]	0.171	0.681	-0.089	0.214	41.278	0.079	0.076
	Organic[c]	1.777	0.191	0.332	0.249	33.783		
	Yrs exp. w. cattle	2.615	0.109	0.008	0.005	104.220		
Needs of cows	Rearing system[a]	9.639	**0.003**	**0.526**	0.169	42.732	0.204	0.331
	Highest education[e]	5.214	**0.025**	**-0.108**	0.047	97.735		
	Age	3.127	*0.080*	*0.008*	0.004	99.133		
	Organic[c]	0.080	0.778	0.056	0.199	37.144		
Impatience moving	Rearing system[a]	2.837	*0.099*	*-0.432*	0.257	44.064	0.068	0.256
	Age	1.964	0.164	-0.009	0.006	97.924		
	Organic[c]	4.385	**0.043**	**0.640**	0.306	39.624		
Patience moving	Rearing system[a]	0.440	0.510	0.103	0.156	43.880	0.067	0.273
	Grown up on farm[d]	8.594	**0.004**	**-0.484**	0.165	89.226		
Patience milking	Rearing system[a]	0.964	0.329	0.168	0.171	89	0.052	0.052
	Yrs exp. w. cattle	1.658	0.201	-0.007	0.006	89		
	Grown up which[f]	1.492	0.225	0.194	0.159	89		
Punish milking	Rearing system[a]	1.325	0.255	-0.412	0.358	47.863	0.060	0.377
	Herd size	2.806	0.102	-0.016	0.009	40.593		
	Organic[c]	1.296	0.261	0.491	0.432	45.426		
Use of kickbow	Rearing system[a]	0.195	0.660	-0.183	0.413	51.306	0.022	0.427
	Yrs exp. w. cattle	2.853	*0.095*	*0.015*	0.009	90.191		
Warning before milking clusters	Rearing system[a]	0.278	0.599	0.083	0.157	112	0.002	0.002
Importance of contact	Rearing system[a]	0.915	0.345	0.182	0.190	36.890	0.073	0.262
	Organic[c]	0.348	0.559	0.133	0.225	31.654		
	Herd size	1.132	0.296	-0.005	0.005	30.550		
	Yrs exp. w. cattle	3.329	*0.071*	*-0.008*	0.004	96.344		
Comfort working with cows	Rearing system[a]	0.477	0.494	-0.104	0.151	44.562	0.073	0.300
	Age	4.754	**0.032**	**-0.009**	0.004	96.380		
	Education[b]	4.583	**0.035**	**0.338**	0.158	97.140		
Liking positive contact	Rearing system[a]	5.985	**0.019**	**0.418**	0.171	37.705	0.162	0.208
	Age	2.430	0.122	-0.015	0.010	102.368		
	Yrs exp. w. cattle	0.245	0.622	-0.005	0.010	104.919		
	Grown up on farm[d]	1.587	0.211	-0.317	0.252	104.577		
Liking negative contact	Rearing system[a]	0.967	0.331	-0.221	0.225	44.793	0.060	0.292
	Yrs exp. w. cattle	1.660	0.201	0.009	0.007	98.625		
	Education[b]	0.596	0.442	0.213	0.275	85.739		
	Grown up farm[d]	0.506	0.479	0.204	0.287	85.842		

[a] Rearing system: 0 = ES, 1 = CCC
[b] Education: 0 = no, 1 = yes
[c] Organic: 0 = no, 1 = yes
[d] Grown up on farm: 0 = no, 1 = yes
[e] Highest education: 0 = no formal agricultural education, 1 = formal agricultural education, level: skilled worker, 2 = formal agricultural education, level: master craftsmen, 3 = university degree in agricultural studies
[f] Grown up on which type of farm: 1 = dairy, 2 = beef fattening, 3 = veal, 4 = beef suckler, 5 = farm without cattle

of the 25 CCC farmers in our sample usually fed (additional) colostrum by hand and thus handled the calves. Moreover, almost half of the CCC farms practiced short-time CCC where farmers regularly, mostly twice a day, interacted with the calves during feeding times.

There was also no difference in the cows' AD. This supports our hypothesis that there is no general negative effect of CCC on the AHR but that a relationship comparable to ES cows can be facilitated through sufficient HAI (Waiblinger et al., 2002). For cows, this is even easier than for calves as cows are regularly milked and thus handled. There could be a higher risk for reduced human contact in (whole-day) foster cow systems for cows that are not milked, although a similar risk applies to cattle on farms with an automated milking system (AMS; Wildridge et al., 2020; in our sample: four whole-day foster systems (of which one with AMS) and two ES farms with AMS). Although, to our knowledge, there are no studies that investigated the cattle-human relationship in foster cow systems, farmers confirmed that regular gentle interactions with CCC cows were sufficient to maintain a good relationship, which is in line with knowledge about the dynamics of HARs in general and of human-cattle relationships specifically (Waiblinger et al., 2006; Waiblinger, 2019).

Conversely, CCC heifers had higher ADs than ES heifers. This is in contrast to previous findings from Waiblinger et al., 2020, who found no difference in the ADs of dam-reared compared to ES heifers at an age of one year. However, in the referenced study, both CCC and ES heifers were managed similarly after weaning at an age of 90 days and thus had similar amounts and qualities of human-cattle interactions (Waiblinger et al., 2020). For the heifers in our study, it can be assumed that they had a different intensity, amount and/or quality of HAI that led to higher ADs. One explanation may be the more extensive housing conditions of the CCC heifers in our study: they had more space and greater access to

Table 2
Spearman rank correlation coefficients between the empathy and attitude factors of n = 57 CCC and n = 57 ES stockpeople from 25 CCC and 25 ES farms, respectively.

Table 2		Pos. Characteris-tics cows	Cows' cognitive abilities	Negative attitudes t. cows	Needs of cows	Impatience moving	Patience moving	Patience milking	Punish milking	Use of Kickbow	Warn bf milking clusters	Importance of contact	Comfort working with cows	Liking positive contact	Liking neg. contact
Empathy	r_s	0.359	0.525	-0.149	0.396	-0.229	0.354	0.210	-0.277	-0.159	-0.136	0.536	0.181	0.365	-0.339
	p	**0.000**	**0.000**	0.111	**0.000**	**0.013**	**0.000**	**0.024**	**0.003**	*0.088*	0.145	**0.000**	**0.052**	**0.000**	**0.000**
Pos. Characteristics cows	r_s	–	0.368	-0.306	0.101	-0.064	0.248	0.325	-0.230	-0.068	-0.107	0.374	0.383	0.308	-0.101
	p	–	**0.000**	**0.001**	0.284	0.498	**0.008**	**0.000**	**0.014**	0.475	0.261	**0.000**	**0.000**	**0.001**	0.289
Cows' abilities	r_s	0.368	–	-0.306	0.199	-0.106	0.213	0.412	-0.238	-0.319	0.045	0.393	0.298	0.417	-0.299
	p	**0.000**	–	**0.001**	**0.033**	0.263	**0.023**	**0.000**	**0.011**	**0.001**	0.634	**0.000**	**0.001**	**0.000**	**0.001**
Negative attitudes t. cows	r_s	-0.306	-0.306	–	0.034	0.216	-0.115	-0.169	0.445	0.164	0.209	-0.035	-0.131	-0.052	0.274
	p	**0.001**	**0.001**	–	0.721	**0.022**	0.226	*0.073*	**0.000**	*0.085*	**0.026**	0.710	0.167	0.588	**0.003**
Needs of cows	r_s	0.101	0.199	0.034	–	-0.005	0.022	0.202	-0.045	-0.135	-0.078	0.259	-0.053	0.198	-0.054
	p	0.284	**0.033**	0.721	–	0.954	0.815	**0.032**	0.638	0.157	0.411	**0.006**	0.575	**0.035**	0.570
Impatience moving	r_s	-0.064	-0.106	0.216	-0.005	–	-0.188	-0.201	0.414	0.236	-0.126	-0.157	-0.166	-0.168	0.562
	p	0.498	0.263	**0.022**	0.954	–	**0.046**	**0.033**	**0.000**	**0.012**	0.183	*0.097*	*0.079*	*0.075*	**0.000**
Patience moving	r_s	0.248	0.213	-0.115	0.022	-0.188	–	0.379	-0.227	-0.099	0.119	0.369	0.012	0.255	-0.392
	p	**0.008**	**0.023**	0.226	0.815	**0.046**	–	**0.000**	**0.015**	0.301	0.208	**0.000**	0.898	**0.006**	**0.000**
Patience milking	r_s	0.325	0.412	-0.169	0.202	-0.201	0.379	–	-0.310	-0.150	0.101	0.337	0.278	0.273	-0.261
	p	**0.000**	**0.000**	*0.073*	**0.032**	**0.033**	**0.000**	–	**0.001**	0.114	0.285	**0.000**	**0.003**	**0.003**	**0.005**
Punish milking	r_s	-0.230	-0.238	0.445	-0.045	0.414	-0.227	-0.310	–	0.294	0.043	-0.119	-0.296	-0.134	0.514
	p	**0.014**	**0.011**	**0.000**	0.638	**0.000**	**0.015**	**0.001**	–	**0.002**	0.652	0.210	**0.001**	0.157	**0.000**
Use of Kickbow	r_s	-0.068	-0.319	0.164	-0.135	0.236	-0.099	-0.150	0.294	–	0.067	-0.198	-0.224	-0.263	0.184
	p	0.475	**0.001**	*0.085*	0.157	**0.012**	0.301	0.114	**0.002**	–	0.485	**0.037**	**0.018**	**0.005**	*0.053*
Warn bf. milking clusters	r_s	-0.107	0.045	0.209	-0.078	-0.126	0.119	0.101	0.043	0.067	–	0.092	-0.026	0.009	-0.104
	p	0.261	0.634	**0.026**	0.411	0.183	0.208	0.285	0.652	0.485	–	0.334	0.782	0.924	0.272
Importance of contact	r_s	0.374	0.393	-0.035	0.259	-0.157	0.369	0.337	-0.119	-0.198	0.092	–	0.237	0.523	-0.183
	p	**0.000**	**0.000**	0.710	**0.006**	*0.097*	**0.000**	**0.000**	0.210	**0.037**	0.334	–	**0.012**	**0.000**	*0.052*
Comfort working w. cows	r_s	0.383	0.298	-0.131	-0.053	-0.166	0.012	0.278	-0.296	-0.224	-0.026	0.237	–	0.390	-0.085
	p	**0.000**	**0.001**	0.167	0.575	*0.079*	0.898	**0.003**	**0.001**	**0.018**	0.782	**0.012**	–	**0.000**	0.369
Liking pos. contact	r_s	0.308	0.417	-0.052	0.198	-0.168	0.255	0.273	-0.134	-0.263	0.009	0.523	0.390	–	-0.253
	p	**0.001**	**0.000**	0.588	**0.035**	*0.075*	**0.006**	**0.003**	0.157	**0.005**	0.924	**0.000**	**0.000**	–	**0.007**
Liking neg. contact	r_s	-0.101	-0.299	0.274	-0.054	0.562	-0.392	-0.261	0.514	0.184	-0.104	-0.183	-0.085	-0.253	–
	p	0.289	**0.001**	**0.003**	0.570	**0.000**	**0.000**	**0.005**	**0.000**	*0.053*	0.272	*0.052*	0.369	**0.007**	–

Table 3
Descriptive results of the Avoidance Distance on 25 farms with cow-calf contact (CCC, for heifers: n = 19 farms) and 25 farms with early separation of cow and calf (ES). AD0 = the percentage of heifers or cows that could be touched per farm, AD50 = the percentage heifers, or cows with an AD > 50 cm.

Table 3	Heifers Median, range		Cows Median, range	
	CCC	ES	CCC	ES
AD0	27, 0–90	36, 16 – 92	62, 31–82	62, 23–84
AD Median	20, 0–45	10, 0–40	0, 0–15	0, 0 – 20
AD50	21, 0–50	9, 0–41	0, 0–14	3, 0–31

pasture (Rademann et al., 2025). While these aspects are beneficial for welfare, they make regular (positive) HAIs and thus the formation and maintenance of a good AHR more difficult. Still, this is surprising, as previous studies have shown that the effects of even short periods of handling during weaning on the AHR persisted until eight months of age (Boivin et al., 1992). However, Bertelsen et al. (2023) found an effect of weaning method on the latency to first approach the human in a stationary human test, suggesting that weaning and separation distress in CCC cattle can negatively impact the AHR. Previous studies have shown the importance of regular positive interactions for maintaining or developing a good relationship, particularly for heifers (Breuer et al., 2003; Lürzel et al., 2016). That there is a general negative effect of the "system" CCC on the AHR is unlikely as there was no difference in the AD for both calves and cows who are most influenced by CCC, but the effect likely can be explained by differences in management and contact intensity in heifers.

4.2.2. Associations of attitudes and human-cattle contact with avoidance distances

Calves and heifers had a lower AD on farms where farm managers agreed more on *positive characteristics of cows*. This is in line with Waiblinger et al. (2002): *positive attitudes* was the only general attitude component that was (positively) associated with positive stockperson behaviour, and this with lower AD.

The slight association between attitudes towards behaviour during moving (only cows) and milking and AD in heifers and cows is consistent with previous findings (Waiblinger et al., 2002). However, we expected stronger associations between farmers' attitudes and the cattle's (especially calves' and cows') ADs, mainly for behavioural attitudes and attitudes towards contact to cows due to previous studies (cattle: Waiblinger et al., 2002; Ebinghaus et al., 2018; Calderón-Amor et al., 2020; sheep: Napolitano et al., 2011; goats: Mersmann et al., 2016). For calves, one reason could be that the questionnaire mainly comprised questions for (handling of) cows and less for calves and heifers (see 4.3 limitations). For cows, the high proportion of low ADs and cows that could be touched and thus a low variability can be an explanatory factor. In addition, AD at the feeding place have a somewhat lower validity in measuring the animals' relation with humans due to more confounding factors such as social stress at the feeding place or type of feed provision (Waiblinger et al., 2003; Ebinghaus et al., 2018).

Table 4
Results of the Linear Mixed model with AD of individual calves as target variable. Due to the stronger correlations of farm manager attitudes with AD in the Spearman rank correlation, attitude factors of farm managers were used as predicting variables. Significant ($p \leq 0.05$) p-values are marked in bold.Model: Marginal $R^b = 0.231$; Conditional $R^b = 0.501$.

Independent variables	β	SE	df	t	p
Rearing system[a]	13.492	10.591	40.449	1.274	0.210
Age category[b]	-2.479	5.734	393.757	-0.432	0.666
Test location: feeding place[c]	-29.384	6.873	376.904	-4.275	**0.000**
Test location: single box[c]	-37.805	7.499	392.605	-5.042	**0.000**
Positive characteristics of cows	-18.491	7.888	38.096	-2.344	**0.024**
Warning before milking clusters	2.548	8.215	35.537	0.310	0.758
Importance of contact	1.390	8.507	35.926	0.163	0.871
Comfort working with cows	-3.204	8.921	37.065	-0.359	0.722

[a] Rearing system: 0 = ES, 1 = CCC
[b] Age category: 0 = 0–2 months, 1 = 3–6 months
[c] Relative to test location "in the barn"; there was no statistically significant difference between the test location "feeding place" and "single box" (p = 0.334, β = −8.421).

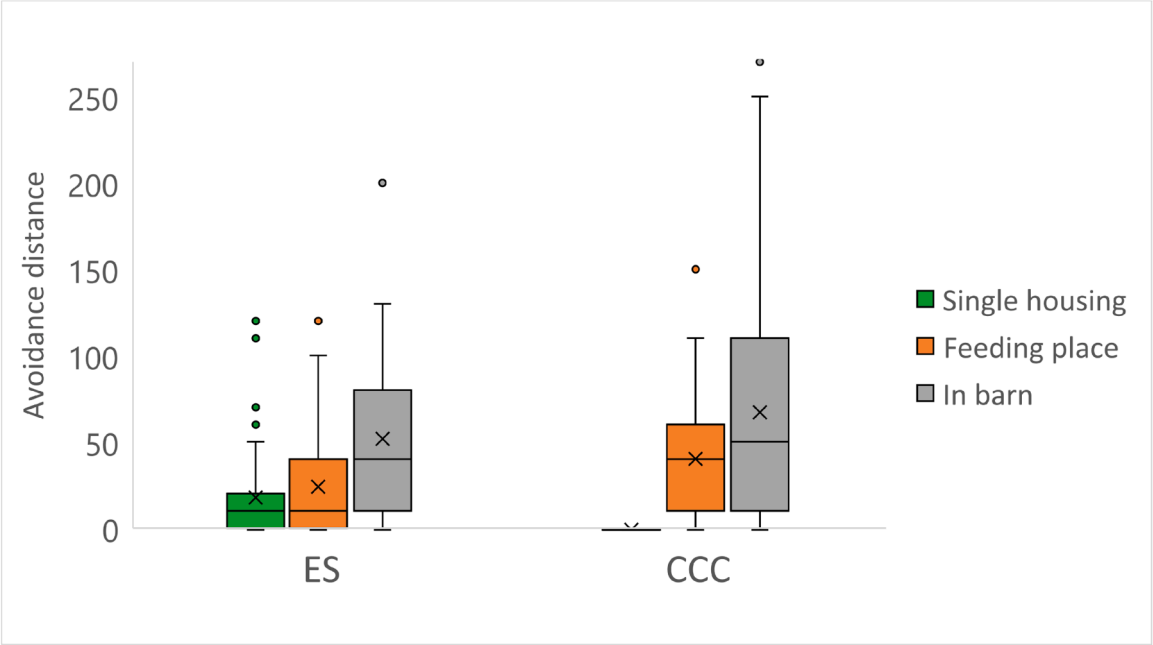


Fig. 2. Avoidance distance of individual calves (CCC: n = 170 calves, ES: n = 233 calves) in the three different test locations single housing (CCC: n = 2, ES: n = 69), feeding place (CCC: n = 41, ES: n = 91), or in barn (CCC: n = 127, ES: n = 73).

Table 5
Results of the linear regression models with AD0, AD Median and AD50 of heifers and cows as target variable. Due to the more and stronger correlations of farm mean attitudes and AD for heifers, farm mean values were used for the attitude factors. Due to the stronger correlations of farm manager attitudes with AD of cows, these values were used for the models for cows. Significant ($p \leq 0.05$) p-values are marked in bold, tendencies ($0.05 < p \leq 0.1$) are marked in *italics*.

Target vari-able; Model: R^b / F / p / n	AD0_heifers			AD Median_heifers			AD50_heifers			AD0_cows			AD Median_cows			AD50_cows		
	0.564 / 3.344 / 0.003 / 44			0.564 / 3.337 / 0.003 / 44			0.557 / 3.246 / 0.004 / 44			0.160 / 0.976 / 0.468 / 50			0.255 / 1.751 / 0.116 / 50			0.210 / 1.361 / 0.242 / 50		
Predicting variable	B	SE	p	B	SE	p	B	SE	p	B	SE	p	B	SE	p	B	SE	p
Rearing system[a]	-15.06	5.04	**0.005**	11.36	3.42	**0.002**	12.76	4.000	**0.003**	2.192	4.398	0.621	-0.859	1.514	0.574	0.152	1.811	0.934
N°. milkers[b]	0.97	3.33	0.772	3.13	2.26	0.175	-0.49	2.643	0.854	0.690	3.219	0.831	0.191	1.108	0.864	0.148	1.325	0.912
Brushing of cows[3]	6.23	4.58	0.183	1.17	3.10	0.709	5.97	3.630	0.110	-2.679	3.980	0.505	-0.659	1.370	0.633	-0.993	1.638	0.548
Time in barn / d	5.64	2.46	**0.029**	-1.00	1.67	0.555	-2.05	1.951	0.301	3.032	1.747	*0.090*	-1.289	0.601	**0.038**	-1.559	0.719	**0.036**
N°. heifers / herdsize	-0.75	0.28	**0.011**	0.70	0.19	**0.001**	0.82	0.221	**0.001**	-0.079	0.163	0.629	-0.014	0.056	0.803	0.069	0.067	0.309
Pos. Characteristics cows	2.33	6.93	0.739	-10.20	4.70	**0.038**	-8.93	5.501	0.114	-	-	-	-	-	-	-	-	
Neg. attitudes cows	-2.02	5.06	0.692	-0.42	3.43	0.903	6.24	4.017	0.130	-	-	-	-	-	-	-	-	
Impatience moving	-4.39	3.42	0.209	0.71	2.32	0.760	1.51	2.716	0.582	-4.214	2.196	*0.062*	1.307	0.756	*0.091*	1.370	0.904	0.137
Patience milking	4.15	5.16	0.427	-3.15	3.50	0.375	-3.32	4.092	0.423	-	-	-	-	-	-	-	-	-
Punish milking	-4.64	2.92	0.122	3.40	1.98	*0.096*	-2.01	2.318	0.393	-	-	-	-	-	-	-	-	-
Importance contact	-3.51	4.64	0.455	-1.29	3.14	0.684	-1.58	3.680	0.670	-	-	-	-	-	-	-	-	-
Comfort working w. cows	2.35	5.21	0.655	0.65	3.53	0.856	-1.84	4.136	0.659	-	-	-	-	-	-	-	-	-
Use of Kickbow	-	-	-	-	-	-	-	-	-	0.286	1.479	0.847	0.684	0.509	0.186	0.274	0.609	0.655
Warn bef. Milk. clusters	-	-	-	-	-	-	-	-	-	-3.354	3.189	0.299	1.474	1.098	0.187	0.760	1.313	0.566

[a] Rearing system: 0 = ES, 1 = CCC; [2]N°. milkers: 0 = 2 or more or robot, 1 = 2, 2 = 1;
[b] Brushing of cows: 0 = maximum 1x / year or never, 1 = less than every month but more than once a year, 2 = at least once a month, following Waiblinger, (1996) Discussion

4.2.3. Other influences on the Avoidance Distance

That heifers and cows on farms where farmers spent more time in the barn had lower ADs is probably due to their habituation with human contact (Waiblinger et al., 2002, 2006). It is also possible that the farmers on these farms consciously took more time for (positive) contact with the cattle, although there was no association with any (behavioural) attitude component.

In addition, the number of heifers was positively associated with AD of heifers in the model, number of cows or calves each were associated with a higher proportion of cows or calves with AD50 in the univariate analysis but not the model. As human-heifer interactions must be facilitated more actively (see 4.2.1), the higher the number of heifers, the more difficult it may be, or the more time may be needed to maintain a good AHR with all of them. For cows, a larger herd size can be a risk factor for reduced human-cattle contact but good HARs are possible in larger herds, as this mainly depends on farmers' personalities and attitudes (Waiblinger and Menke, 1999; Waiblinger et al., 2025).

4.3. Practical application and future research

Attitudes, norms, motivation and capability shape farmer behaviour in livestock management that, in turn, leads to improved livestock welfare. Our findings suggest an association between attitudes, especially towards needs of cows, and the choice of rearing system. This means that, for practical application, *i.e.,* the up-scaling of CCC systems in practice, also farmers' attitudes should be targeted in addition to conventional ways of distributing information and arranging circumstances that make adoption of CCC easier. Many theories, mainly developed to address health-related behaviours, are relevant to human behaviour in managing and interacting with livestock. Such theories help to identify the relevant human variables that need to be targeted (Hagger et al., 2020). These variables can be considered as a hierarchy with social structures and constraints through to values, attitudes and norms to behaviour. Empirical evidence from experimental, on-farm, and post-farm-gate studies demonstrates the successes and limitations of current interventions. Stockperson training programmes that have utilised experimental and on-farm research have been developed for use on-farm in the pork and dairy industries and have demonstrated effectiveness in bringing about behaviour change leading to improved animal welfare (Hemsworth et al., 1994, 2002). As attitudes are learned dispositions, they can be changed, and previous studies revealed a high effectiveness of combined cognitive-behavioural interventions to change attitudes and improve handling of stockpeople (Hemsworth et al., 1994, 2002; reviewed in Coleman and Hemsworth, 2014). Among personality, social pressure, the strength of current attitudes, the source (s) of information and the characteristics of the information (e.g. order of arguments, type of appeal) affect attitudes and thus behaviour change, so these aspects should be taken into consideration in targeted interventions (reviewed in Hemsworth and Coleman, 2011). Specifically, information about cows' and calves' natural behaviour, including maternal behaviour and cow-calf interactions and the effects of CCC on welfare and behaviour, along with best practices of handling cows and calves together and individually (= cognitive) should be combined with practical, on-farm experience (= behavioural). Generally, recommendations for a positive HCR (reviewed in Waiblinger and Lürzel, 2023) should be conveyed, and emphasis be laid on sensitive periods (Boivin et al., 1992; Waiblinger et al., 2020) and on the importance of regular contact to CCC heifers specifically, and extensively housed cattle in general (Waiblinger et al., 2002; Lürzel et al., 2015).

It would be interesting for future studies to design such a cognitive-behavioural intervention program targeted at dairy farmers to change from ES to a CCC system and to investigate its effectiveness in relation to changes of attitudes and actual change of system. Moreover, to better understand the effects of and on different CCC systems, attitudes of farmers of different CCC systems (dam vs. foster cow, whole-day vs. part-time) and their association with cattle welfare, job satisfaction and

other sustainability aspects should be investigated, which was not possible in our study due to the small sample size per CCC system.

4.4. Limitations

The questionnaire on attitudes primarily contained questions relating to cows. Only few sections contained items on calves or heifers, mainly in the component *importance of contact*, or some on cattle more generally. Because highest association with behaviour can be expected with specific behavioural attitudes (Ajzen and Fishbein, 1980; Ajzen, 2011), stronger associations could be expected for calves and heifers if the questionnaires were tailored more specifically. In addition, the study is limited by the convenience sampling and self-selection of the farms. While the CCC farms are likely a representative sample, ES farms are likely to be subjected to a positive bias due to the sampling procedure. Moreover, the nature of the questions and the familiarity with the researchers, resulting from the farm visit, might have been introduced social desirability bias, which could have led to more positive results (Joinson, 1999; Rickwood and Coleman-Rose, 2023).

5. Conclusion

The results confirm the hypothesis of more positive attitudes of CCC compared to ES farmers, and strong correlations between attitudes and empathy. However, no general effect of CCC on the cattle's relationship to humans was confirmed; only in heifers a difference was found with CCC heifers having higher ADs compared to ES heifers. In addition, the results confirm an influence of human attitudes on the cattle's relationship to humans.

CRediT authorship contribution statement

Anna Rademann: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Marie Louise Schneider:** Writing – review & editing, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Andreas Futschik:** Writing – review & editing, Formal analysis. **Grahame Coleman:** Writing – review & editing, Methodology. **Susanne Waiblinger:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.applanim.2026.107187.

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